Created in Nepal, September 2013

Management of rice leaf folder

Recognize the problem

The adult rice leaf folder is a yellow-brown moth. Leaf folder caterpillars fold the rice leaves around themselves and attach to the leaf margins. Each female lays around 300 eggs at night, during a lifecycle of 3 to 10 days. Larvae are yellow and about 12 - 25 mm long. As they mature, they turn yellowish green with brown heads. Each larva can make two to four folded leaves. The pupa is about 9-12 mm long and is found inside the rolled leaf. Adults are yellowish brown in colour and usually emerge in the evening.

Leaf folder larva (Photo by Source: IRRI (CC BY-NC-SA))



Adult (Photo by Luvjoy Choker (CC BY-NC))



Background

Folded leaves restrict photosynthesis while harbouring inside the folded leaf. The caterpillar feeds by scraping off the leaf surface tissue. The damage looks bad, but rarely reduces yields. Rice plants are susceptible to leaf folders from seedling to flowering. Before panicle initiation, 50% of the leaves can be damaged without reducing yields. However, if the leaf folder damages the flag leaf (i.e. the last leaf to emerge from the plant), yield can be greatly reduced. The factors favouring insect development are: heavily fertilized fields, high humidity, shady areas and high plant density.

Management

- Natural enemies like parasitic wasps, predatory beetles, spiders, and predatory crickets may control leaf folders. Killing biological control agents with pesticides could lead to pest outbreaks. Therefore, it is recommended that insecticides are not used haphazardly.
- The cultural practices that can prevent leaf folder are crop rotation, avoidance of ratoon plants, removal of grassy weeds from the field and borders, application of standard plant density and the use of a balanced dose of fertilizer.
- Monitor the paddy field carefully. Destroy the eggs observed in the plants. Commonly, eggs are seen at the tip of the leaves. Break the silk strands that hold the edges of folded leaves and remove the caterpillars inside. Do not drop the live caterpillars into the paddy water as they may find their way back to the rice plants.
- Even when maintaining the above mentioned techniques, if the population of insects is increasing, spray a chemical insecticide like malathion or chlorpyriphos at rate of 5 ml per litre of water or cypermethrin at rate of 1.5 ml per litre of water

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > **Cnaphalocrocis medinalis**

The recommendations in this factsheet are relevant to: Nepal

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Edited by Plantwise.

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